

1 Updated Work Plan

Since the project started in September 2025, it has progressed well, and the scientific direction has become clearer. The overall objectives and the five work packages remain unchanged. However, the team structure is being updated to better support the project's current needs.

The original staffing plan consisted of one four-year postdoctoral researcher and one PhD student. The revised structure consists of one two-year postdoctoral researcher and two PhD students. This provides a better balance between the experimental work and the increasingly important theoretical and computational aspects of the project, while maintaining the original scientific objectives.

The first PhD student joined in November 2025 and is currently building experimental expertise while contributing to the project's early experimental implementation. A second PhD student is expected to join in September 2026 and will focus on the theoretical and computational aspects of TRAD, including computational modelling, simulations and the development of automated design tools. The postdoctoral researcher is also expected to start in September 2026 and will work across both the experimental and computational activities during the first two years, helping establish the experimental platform while ensuring close interaction between modelling and experimental validation.

Recruitment of the postdoctoral researcher has taken longer than originally anticipated. Two previous recruitment rounds did not result in an appointment. The first did not identify a suitable candidate, while the second resulted in an offer that was later declined due to personal and relocation reasons. A third recruitment round is currently underway, and interviews are taking place.

The budget has been reprofiled to reflect the revised team structure. While the scientific objectives remain unchanged, the work packages have been updated to reflect the revised distribution of responsibilities and the closer interaction between computational modelling and experimental implementation throughout the project.

The work plan continues to comprise five work packages that address the project's four objectives.

WP#	Work Package Title	Associated Objective
WP1	Theoretical Development and Design.	O1, O2
WP2	Experimental Setup and Protocol Development.	O3
WP3	Sustainable Molecular Computing	O4
WP4	Dissemination and Communication.	All Os
WP5	Management and Training.	All Os

1.1 WP1: Theoretical Development and Design (months 1-24)

Lead: PI with PhD2, supported by PD

Associated Objectives:

- **O1.** Validate a DNA Computing Platform with Thermodynamically Favoured Outputs
- **O2.** Develop a Library of Thermally-Regulatable DNA Motifs

This work package develops the computational foundations of TRAD. The focus is on theoretical modelling, thermodynamic analysis, sequence design, computational optimisation and simulation of thermally regulated DNA motifs. Rather than being completed before the experimental work begins, WP1 will progress alongside WP2, with computational predictions continuously refined using experimental observations.

Objective O1 will validate the thermodynamic behaviour of the Scaffolded DNA Computer and establish the theoretical framework required for thermally regulated computation.

Objective O2 will develop and computationally optimise a library of thermally regulatable DNA motifs. This includes developing automated modelling workflows in NUPACK, sequence optimisation, simulation tools and computational methods for analysing motif behaviour under different temperature programmes.

The second PhD student will lead much of the computational development under the PI's supervision, while the PD ensures that computational predictions remain closely aligned with experimental implementation.

Deliverables: **D1.1** Literature review in an open science journal to allow open access for researchers interested in the topic. Manuscript submission #1 is a literature review on renewable DNA computation methods to date. (*month 6*). **D1.2** A library of thermally regulatable DNA hairpin-based motifs. (*month 18*). **D1.3** TFC temperature programming model. (*month 24*). **D1.4** Manuscript submission #2 is a theory paper

with a modelling library in a field-specific annual DNA conference, like the annual International Conference on DNA Computing and Molecular Programming (*month 24*).

1.2 WP2: Experimental Setup and Protocol Development (months 25-36)

Lead: PD and PhD1

Associated Objective:

- **O3.** Create an Automated One-Pot System for Molecular Computing

This work package develops the experimental implementation of TRAD. Experimental work will progress iteratively alongside the computational developments in WP1. As computational models mature, new motif designs will be synthesised, experimentally characterised and fed back into the computational framework for further optimisation.

PhD1 will lead the experimental work, while the PD will establish laboratory protocols, optimise experimental procedures, and coordinate the integration of modelling with experimental validation.

The work focuses on the validation of thermally regulated motifs, implementation on the Scaffolded DNA Computer, optimisation of protocols and development of an automated one-pot computing system.

Deliverables: **D2.1** Manuscript submission #3 is an experimental paper in *Nature Communications* or a similar journal (*month 36*). **D2.2** A prototype of TRAD on SDC.

1.3 WP3: Sustainable Molecular Computing (months 31-48)

Lead: PhD1 (experiments) and PhD2 (modelling), supported by the PI

Associated Objective:

- **O4.** Enable Energy-Efficient and Sustainable Molecular Computing Applications.

This work package brings together the computational framework developed in WP1 and the experimental platform established in WP2 to further develop the TRAD approach beyond the initial proof of concept. The aim is to develop a robust and scalable framework for thermodynamically regulated molecular computing that can be adopted for a wider range of DNA computing systems.

PhD1 will lead the experimental validation of new motif designs and their integration into the Scaffolded DNA Computer, while PhD2 will lead the computational modelling, optimisation and development of additional motif libraries. The two activities will continuously inform each other, allowing experimental observations to refine computational models and enabling rapid experimental evaluation of new computational designs. This WP will deliver an integrated computational and experimental framework for sustainable molecular computing together with an expanded collection of validated thermally regulated DNA motifs.

Deliverables: **D3.1** Manuscript submission #4 is an experimental paper in *Science*, *Nature*, *PNAS* or similar (*month 48*). **D3.2** A dataset of DNA sequences of thermally regulatable motifs (*month 48*).

1.4 WP4: Dissemination and Communication (months 7-48)

Lead: PI / Associated Objectives: All

Dissemination will continue throughout the project to maximise both scientific impact and public engagement. Results will be communicated through high-impact journals, international conferences, invited talks, workshops and outreach activities. Particular emphasis will be placed on engaging both the DNA computing community and the wider molecular programming and computer science communities.

Project outputs will be presented regularly at leading international conferences, including the International Conference on DNA Computing and Molecular Programming (DNA), Foundations of Nanoscience (FNANO), IEEE NANO and other relevant meetings. Early dissemination has already begun, with a poster presented at Maynooth University's Faculty of Science and Engineering Research Connections event in May 2026, and a second poster accepted for presentation at DNA32 in the US in August 2026.

Public engagement activities also began in late 2025 through Maynooth University's Science Night and a Women & Girls in STEM event, reaching more than 130 children, families and young women through hands-on DNA computing activities. These activities will continue throughout the project, using accessible and gender-inclusive communication where appropriate.

Deliverables: **D4.1** Invited talk to present the theory of TRAD in a field-relevant DNA conference, *DNAx*, *FNANO* or similar. **D4.2** Publication of manuscripts from WP1–WP3. **D4.3** Two PhD theses. **D4.5** Article for the public in RTE Brainstorm or similar

1.5 WP5: Management and Training (months 1-48)

Lead: PI / Associated Objectives: All

This WP aims to further develop the PI's project management and leadership skills, support the PD's career development, and enhance the research skills of both PhD students.

Deliverables: **D5.1** Career development plan for the PD. **D5.2** Data management plan. **D5.3** PhD students to participate in Maynooth University's Research Skills Development Programme (RSDP). **D5.4** Invited talks and organising workshops. **D5.5** Further grant submission, *e.g.* ERC Consolidator, to obtain funding to establish an independent research group.

Timing of work plan: Gantt Chart. All PI, PD, PhD1, and PhD2 will contribute to all deliverables. **Only the lead** is shown in the deliverables' cells.

Activity	Start month	End month	Year 1				Year 2				Year 3				Year 4			
			1	4	7	10	13	16	19	22	25	28	31	34	37	40	43	46
WP1	1	24	PI	PI	PI	PI	PI	PI	PI	PI								
D1.1	1	6	PhD1	PhD1														
D1.2	1	18			PhD1	PhD1	PhD2	PhD2										
D1.3	19	24							PhD2	PhD2								
D1.4	4	24					PD	PD	PD	PD								
WP2	25	36									PI	PI	PI	PI				
D2.1	25	36									PhD2	PhD2	PhD1	PhD1				
D2.2	25	36									PD	PD	PD	PD				
WP3	31	48											PD	PD	PI	PI	PI	PI
D3.1	31	48											PhD1	PhD1	PhD1	PhD1	PhD1	PhD1
D3.2	31	48											PhD2	PhD2	PhD2	PhD2	PhD2	PhD2
WP4	7	48			PI	PI	PI	PI	PI	PI	PI	PI	PI	PI	PI	PI	PI	PI
WP5	1	48	PI	PI	PI	PI	PI	PI	PI	PI	PI	PI	PI	PI	PI	PI	PI	PI

2 Conclusion, potential risks, and mitigation

The revised work plan maintains the original scientific objectives of TRAD while updating the staffing structure to better reflect the project's current needs. The revised balance between experimental and computational research is expected to advance thermodynamically regulated DNA computing and provide a more integrated workflow throughout the project.

The main remaining risk is recruitment. While the first PhD student is already in place, the project depends on the timely recruitment of the second PhD student and the postdoctoral researcher. Recruitment of the postdoctoral researcher has taken longer than anticipated, but a new recruitment round is currently underway, and interviews are taking place. The second PhD student has been identified and is expected to start in September 2026.

From a scientific perspective, the main technical risks relate to the design and experimental validation of thermally regulated DNA motifs. These risks are mitigated by the close interaction between computational modelling and experimental implementation, enabling iterative design refinement before large-scale experimental validation.

The revised team structure provides complementary expertise across modelling, software development and experimental DNA nanotechnology, further increasing the likelihood of achieving the project's objectives while maintaining flexibility to respond to new scientific findings.